深度學習理論與應用

(Deep Learning Theory and Applications)

Course number：MEME552
Time：9:10-12:00 am, Tuesday
Location：工 EN 4055-2

Wei-Chih Tseng (曾偉智)

E-mail: wctseng@mail.nsysu.edu.tw

Final Project Information

- **Date:** June 2nd and June 9th, 2026
  - TA will randomly assign the date of your group after registration.
- **Group size:** 3 students / group
- **Registration:** Register to TA before April 28th
- **Format:** Oral presentation in English
- **Duration:** 10-15 minutes presentation & 3 minutes Q&A

AI Applications in Smart Factory

- The following are three major categories of smart factory applications. Please select one as your core theme for your final project. Use your creativity to design a specific task and build a deep learning architecture to solve the challenges.

- **Quality Control & Defect Detection Systems**

Ex:
- Pass or Fail detection of the products.
- Automatically scratch the defects on the products/surface.

⋮

AI Applications in Smart Factory

Predictive Maintenance for Machinery

Ex:
- Predict the health of the equipment.
- Predict if a system or process is about to fail.

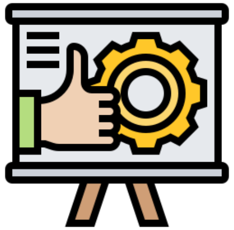
⋮

Sorting & Grasping Guided Systems

Ex:
- Identification and count of screws, gears, or tools.
- Object and position detection for guided grasping.

⋮

Be creative but practical !



Submission & Slides Requirements

- **Submit a single .zip file that includes:**
  - All data sources
  - Source code (.ipynb format)
  - Presentation slides

- **Presentation & slides requirements:**
  - ✓ Introduce the task and define the challenges.
  - ✓ Explain the data sources and preprocessing steps.
  - ✓ Explain your model architecture.
  - ✓ Demonstrate your work and show the final prediction results.

Grading Criteria on Final Project

- Problem Value & Creativity **(20%)**
- Technical challenge **(20%)**

Ex: ✓ Custom / open-source datasets,
✓ Single / multi-objectives
⋮

- Architecture & Completeness **(20%)**

Ex: ✓ Model sophistication,
✓ Logical architecture & pipeline
⋮

★☆☆ CNN classification
★☆☆ RNN regression
★★☆ Object detection
★★★ Video / Hybrid model

- Prediction results **(20%)**

Ex: ✓ Accuracy on test data

- Oral presentation **(20%)**